Sample Paper 08

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

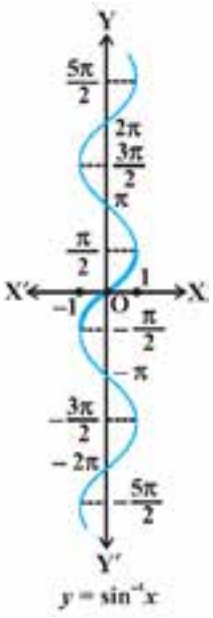
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What are the extreme points on the X-axis visible in the graph of $y = \sin^{-1}x$?



(a) $-\pi$ and π (b) $-\pi/2$ and $\pi/2$

(c) –1 and 1 (d) –2 and 2

2. The area of enclosed by $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$ is

(a) 12 sq units

(b) 13 sq units

(c) $13\frac{1}{2}$ sq units

(d) 14 sq units

3. The general solution of the differential equation $x(1 + y^2)\,dx + y(1 + x^2)\,dy = 0$ is

(a) $(1 + x^2)(1 + y^2) = 0$

(b) $(1 + x^2)(1 + y^2) = c$

(c) $(1 + y^2) = c(1 + x^2)$

(d) $(1 + x^2) = c(1 + y^2)$

4. The foot of the perpendicular from (0, 2, 3) to the line $\frac{x+3}{5} = \frac{y=1}{2} = \frac{z+4}{3}$ is

(a) $(-2, 3, 4)$ (b) $(2, -1, 3)$

(c) $(2, 3, -1)$ (d) $(3, 2, -1)$

5. If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.3$, then $P\left(\frac{A'}{B}\right)$ is equal to

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

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6. The value of $\sin\left[2\cos^{-1}\left(-\frac{3}{5}\right)\right]$ is

(a) $\frac{24}{25}$ (b) $-\frac{24}{25}$

(c) $\frac{7}{25}$ (d) none of these

7. $\int \frac{\sin 2x}{\sin^2 x + 2\cos^2 x}dx$ is equal to

(a) $-\log(1+\sin^2 x)+C$

(b) $\log(1+\cos^2 x)+C$

(c) $-\log(1+\cos^2 x)+C$

(d) $\log(1+\tan^2 x)+C$

8. $\int_0^{\pi}\sqrt{\frac{1+\cos 2x}{2}}dx$ is equal to

(a) 0 (b) 2

(c) 4 (d) -2

9. If $AB = A$ and $BA = B$, then B^2 is equal to

(a) B (b) A

(c) $-B$ (d) B^3

10. The solution of $e^{dy/dx} = x+1, y(0)=3$, is

(a) $y = x\log x - x + 2$

(b) $y = (x+1)\log(x+1) - x + 3$

(c) $y = (x+1)\log(x+1) + x + 3$

(d) $y = x\log x + x + 3$

11. $(x^2+xy)\,dy = (x^2+y^2)\,dx$ is

(a) $\log x = \log(x-y) + \frac{y}{x} + c$

(b) $\log x = 2\log(x-y) + \frac{y}{x} + c$

(c) $\log x = \log(x-y) + \frac{x}{y} + c$

(d) none of the above

12. If $\vec{a}$ and $\vec{b}$ are position vectors of A and B respectively, then the position vector of a point C in AB produced such that $\overrightarrow{AC} = 3\overrightarrow{AB}$, is

(a) $3\vec{a} - \vec{b}$ (b) $3\vec{b} - \vec{a}$

(c) $3\vec{a} - 2\vec{b}$ (d) $3\vec{b} - 2\vec{a}$

13. If $\vec{a} = \hat{i}+\hat{j}, \vec{b} = 2\hat{j} - \hat{k}$ and $\vec{r}\times\vec{a} = \vec{b}\times\vec{a}$, $\vec{r}\times\vec{b} = \vec{a}\times\vec{b}$, then $\frac{\vec{r}}{|\vec{r}|}$ is equal to

(a) $\frac{1}{\sqrt{11}}(\hat{i}+3\hat{j}-\hat{k})$

(b) $\frac{1}{\sqrt{11}}(\hat{i}-3\hat{j}+\hat{k})$

(c) $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$

(d) none of these

14. A mapping $f: n \to N$, where N is the set of natural numbers is define as

$$f(n) = \begin{cases} n^2, & \text{for } n \text{ odd} \\ 2n+1, & \text{for } n \text{ even} \end{cases}$$

for $n \in N$. Then, f is

(a) Surjective but not injective

(b) Injective but not surjective

(c) Bijective

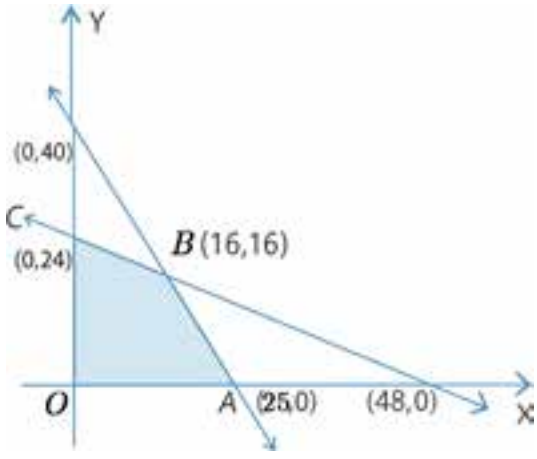
(d) neither injective nor surjective

15. The function $f(x) = x^2 - 2x$ is increasing in the interval

(a) $x \neq -1$ (b) $x \geq -1$

(c) $x \neq 1$ (d) $x \geq 1$

16. Using the same feasible region as shown in the figure, determine the minimum value of the objective function $z = 5x + 2y$. The feasible region has vertices at O(0,0),A(25,0),B(16,16), and C(0,24). At which point is this minimum value attained, and what is it equal to?



(a) 50 (b) 80

(c) 100 (d) 0

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17. In the given figure, the shaded region is bounded by which of the following?

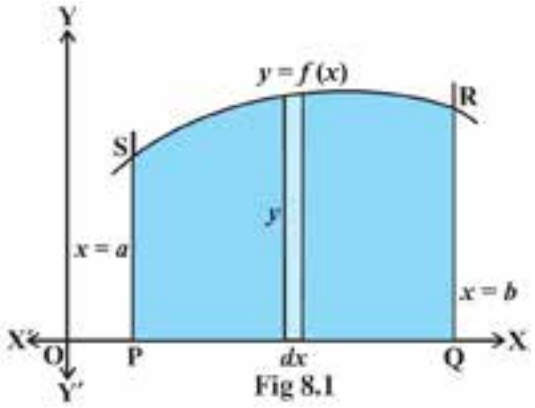


Fig 8.1

(a) y=c, y=d, the curve x=g(y), and the Y-axis

(b) x=c, x=d, the curve y=g(x), and the X-axis

(c) y=0, y=d, the curve x=g(y), and the X-axis

(d) x=0, x=d, the curve y=g(x), and the Y-axis

18. If $R=\{(3,3),(6,6),(9,9),(12,12),(6,12),(3,9),(3,12),(3,6)\}$ is a relation on the set $A=\{3,6,9,12\}$. Then, the relation is

(a) An equivalence relation

(b) Reflexive and symmetric

(c) Reflexive and symmetric

(d) Only reflexive

19. For any square matrix A with real number entries consider the following statements.

Assertion: $A+A'$ is a symmetric matrix.

Reason: $A-A'$ is a skew-symmetric matrix.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. Let A and B be two events associated with an experiment such that $P(A\cap B)=P(A)P(B)$

Assertion: $P(A\mid B)=P(A)$ and $P(B\mid A)=P(B)$

Reason: $P(A\cup B)=P(A)+P(B)$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. Find $|\vec{x}|$, if for a unit vector $\hat{a}$, $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=15$.

22. If $A=\{1,2,3\}, B=\{4,5,6,7\}$ and $f=\{(1,4),(2,5),(3,6)\}$ is a function from A to B. State whether f is one&one or not.

23. It is given that the events A and B are such that $P(A)=\frac{1}{4}$, $P(\frac{A}{B})=\frac{1}{2}$ and $P(\frac{B}{A})=\frac{2}{3}$. Then, find $P(B)$?

OR

Find the probability distribution of X, the number of heads in a simultaneous toss of two coins.

24. Find the vector equation of the line passing through the point $A(1,2,-1)$ and parallel to the line $5x-25=14-7y=35z$.

25. Evaluate $\int (ax-b)^3\,dx$.

OR

Evaluate $\int \frac{(1+\log x)^2}{x}\,dx$.

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. Find the maximum value of $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cos\theta \end{vmatrix}$

27. Show that the function $f(x)=x^3-3x^2+3x$, $x\in R$ is increasing on R.

OR

Find the intervals in which the function

$f(x)=\frac{3}{2}x^4-4x^3-45x^2+51$ is

(i) strictly increasing

(ii) strictly decreasing.

28. Let $\vec{a}=4\hat{i}+5\hat{j}-\hat{k}$, $\vec{b}=\hat{i}-4\hat{j}+5\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}-\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{c}$ and $\vec{b}$ and $\vec{d}\cdot\vec{a}=21$.

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29. Evaluate $\int \frac{x\cos^{-1}x}{\sqrt{1-x^2}}\,dx$.

OR

Evaluate $\int \frac{\sin^6 x+\cos^6 x}{\sin^2 x\cos^2 x}\,dx$.

30. Write the value of $\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$.

31. If $(x-y)\,e^{\frac{x}{x-y}}=a$, prove that $y\frac{dy}{dx}+x=2y$.

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. Find the particular solution of the differential equation $\log\left(\frac{dy}{dx}\right)=3x+4y$, given that $y=0$, when $x=0$.

OR

Find the particular solution of the differential equation $x(1+y^2)\,dx-y(1+x^2)\,dy=0$, given that $y=1$, when $x=0$.

33. Maximize $Z=3x+2y$ subject to $x+2y\le 0$, $3x+y\le 15$, x, $y\ge 0$.

OR

Minimise $Z=x+2y$ subject to $2x+y\ge 3$, $x+2y\ge 6$, x, $y\ge 0$. Show that the minimum of Z occurs at more than two points.

34. If $A=\begin{bmatrix}2 & 3 & 10\\ 4 & -6 & 5\\ 6 & 9 & -20\end{bmatrix}$, find A^{-1}.

Using A^{-1} solve the system of equations $\frac{2}{x}+\frac{3}{y}+\frac{10}{z}=2$, $\frac{4}{x}-\frac{6}{y}+\frac{5}{z}=5$ and $\frac{6}{x}+\frac{9}{y}-\frac{20}{z}=-4$

35. Show that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ intersect. Also, find their point intersection.

SECTION - E

Case study based questions are compulsory.

36. Bob is taking a learning test in which the time he takes to memorize items from a given list is recorded. Let $M(t)$ be the number of items he can memorize in t minutes. His learning rate is found to be

$$M'(t)=0.4t-0.005t^2$$



(i) How many items can Bob memorize during the first 10 minutes?

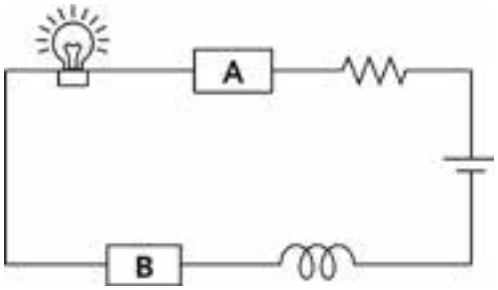
(ii) How many additional items can he memorize during the next 10 minutes (from time $t=10$ to $t=20$)?

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37. An electro-mechanical assembler basically makes machines or/and other assemblies that contain electronic components like wires or microchips. Typically assemblers use blueprints, work instructions, and computer software to manufacture whatever they are working on.



An electronic assembly consists of two sub-systems say A and B as shown below.



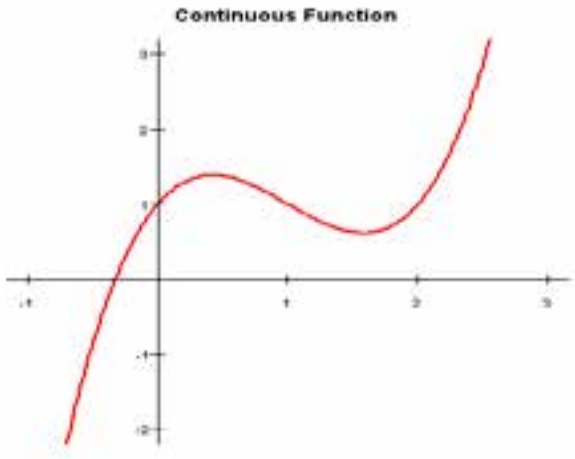
From previous testing procedures, the following probabilities are assumed to be known $P(A \text{ fails}) = 0.2$, $P(B \text{ fails alone}) = 0.15$, $P(A \text{ and } B \textit{ fail}) = 0.15$.

On the basis of above information, answer the following questions.

(i) Find the probability P(B fails) and the probability P(A fails alone).

(ii) Find the probability P(whole system fail) and the probability P(A fails/B has failed).

38. In mathematics, a continuous function is a function such that a continuous variation of the argument induces a continuous variation of the value of the function. This means that there are no abrupt changes in value, known as discontinuities.



Let $f(x)$ be a continuous function defined on $[a, b]$, then

$$\int_a^b f(x)\,dx = \int_0^b f(a+b-x)\,dx$$

On the basis of above information, answer the following questions.

(i) Evaluate $\int_1^2 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{3-x}}\,dx$

(ii) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log \tan x\,dx$

iii) Evaluate $\int_a^b \frac{x^{\frac{1}{n}}}{x^{\frac{1}{n}}+(a+b-x)^{\frac{1}{n}}}\,dx]$

OR

Evaluate $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}\,dx$

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